

CE 520: Environmental Reaction Modeling

Course Code	CE 520
Course Name	Environmental Reaction Modeling
L-T-P-C	3-0-0-3
Prerequisites	IC230, CE519 or similar
Intended for	PG, PhD
One-time approval	54th BoA (May 2024), in the B.Tech Civil Engineering curriculum; 57th BoA (Oct 2024)

Course Contents

- **Modeling basics:** This chapter introduces fundamentals of environmental modeling. First and the most critical step in developing environmental models is conceptualizing the system or process of interest in a useful manner; introduction to various environmental systems and the processes therein that can be quantitatively modeled; concepts of mass and energy transfer, thermodynamics. Environmental engineers and scientists now use quantitative models to understand sediment determine which contaminants will migrate from mine tailings and toxic waste sites, predict scaling in geothermal wells and the outcome of steam-flooding oil reservoirs, solve kinetic rate equations, manage injection wells, evaluate laboratory experiments, and study acid rain. (8 hours).
- **Equilibrium of natural waters:** Chemistry of natural waters is of extreme importance for environmental scientists and engineers in terms of determining the quality of natural waters as well as for the treatment of contaminated water systems. This section provides insights on several steps in environmental modeling. (14 hours).
 - Equilibrium State. This chapter discusses how can we express the equilibrium state of such a system; a direct approach to write each reaction that could occur among the system's species, minerals, and gases; numerical solution for equilibrium state by determining a set of concentrations that simultaneously satisfy the mass action equation corresponding to each possible reaction.
 - Solving for equilibrium state. The principal unknowns equations governing the equilibrium state of aqueous and solid environmental systems are the mass of water and concentrations of basic species, and moles of minerals in equilibrium. Many of these equations are often non-linear and hence cannot be solved using linear algebra. This chapter will focus on the special difficulties posed by the nonlinear forms of the governing equations and discuss how the Newton–Raphson method can be used in geochemical modeling to solve the equations rapidly and reliably.
 - Setting up a model. The basis in reaction modeling includes water, each mineral in the equilibrium system, and each gas of known fugacity, and certain aqueous species. The basis servers two main purposes, each chemical reaction in the model is written in terms of members of the basis set. This chapter discusses how an environmental system can be expressed in terms of chemical reactions and quantitative parameters in the model.
 - Equilibrium model of natural waters. This chapter will focus on constructing models of natural waters to predict processes like mineral dissolution, aqueous speciation of chemicals, minerals precipitation, gas dissolution and degassing, and pH of natural waters.
 - Redox disequilibrium. Many chemical reactions in the environment are redox (oxidation-reduction) reactions and are extremely important for contaminant mobilization. In this chapter, modeling of redox sensitive elements in the environment (C, N, S, O, Fe, and other toxic metals) will be discussed. Inclusion of redox disequilibrium in chemical reaction modeling advances our abilities to predict behavior of contaminants in the environment.
 - Sorption and ion exchange. An important consideration in constructing environmental chemical models, especially those applied to environmental problems, is to account for the sorption of aqueous species onto sediment surfaces. Because of their large surface areas and high reactivities, many components of a sediment – especially clay minerals, zeolites, metal oxides and oxyhydroxides, and organic matter – can sorb considerable masses. This chapter will focus on several simple models of ion sorption and exchange that can be applied within the context of a environmental reaction model. These models include distribution coefficients, Freundlich and Langmuir isotherms, and ion exchange theory.

- Surface complexation. The sorption models presented in the previous chapter are, however, too simplistic to be incorporated into a geochemical model intended for use under general conditions, such as across a range in pH. This chapter discusses a theory of surface complexation to describe hydrolysis and the mineral surface, account for electrical charge there, and provide for mass balance on the sorbing sites. Several surface complexations models will be constructed to predict the chemistry of natural water in equilibrium with minerals with specific reactive surface area.
- **Reaction Processes:** This section overviews specific reaction processes such as mass transfer, polythermal reactions, geochemical buffering, kinetics of mineral dissolution and precipitation, redox kinetics, microbial kinetics, stable isotopes, transport in flowing groundwater, and reactive transport. (12 hours).
 - Mass transfer. In this chapter we consider how to construct reactions paths that account for the effects of simple reactants, a name given to reactants that are added to or removed from a system at constant rates.
 - Polythermal reactions. This chapter will focus on constructing reaction models that varies over temperature. A temperature varies as a function of reaction progress, is numerically modeled, and activity of solutes, stability of minerals, fugacity of gases involved in the system is calculated as a function of temperature.
 - Geochemical buffers. Buffers are reactions that at least temporarily resist change to some aspect of fluid chemistry (e.g., pH buffers). This chapter focuses on constructing models of buffering reactions, both homogenous and heterogenous.
 - Kinetics of dissolution and precipitation. This chapter focuses on modeling environmental reactions to predict how much time it'd take to reach a certain point along the reaction path, calculating relative rates of minerals in rocks, predicting chemistry of waters in equilibrium with certain rocks for a known amount of time, predicting future groundwater quality under known geochemical conditions, by incorporating reaction rate laws from the field of geochemical kinetics.
 - Redox kinetics. The subject of this chapter is modeling the rates at which redox reactions proceed within the aqueous solution, or when catalyzed on a mineral surface or by the action of an enzyme.
 - Microbial kinetics. This chapter focuses on how the microbial community catalyzes redox reactions, perhaps changing in size and composition as it does. The kinetics of such reactions are of special interest, because of the close relationship between geochemical conditions and microbial ecology. The microbes promote reactions that change geochemical conditions, many times significantly, and the geochemistry controls the nature of the microbial community that can exist in a given environment.
- **Applied Reaction Modeling:** This section overviews specific examples of modeling environmental systems such as geothermal fluids, geothermometry, evaporation, sediment diagenesis, kinetics of water rock interaction, weathering, oxidation and reduction, waste injection wells, petroleum reservoirs, acid mine drainage, contamination and remediation, and microbial communities. These models of these (and other) environmental systems will be constructed throughout the duration of course using modeling programs like PHREEQC, Visual MINTEQ and Geochemist's Workbench. The concepts learnt in previous chapters will be utilized in construction of these models. (8 hours).

Text Books

1. Bethke, C.M., **Geochemical and biogeochemical reaction modeling**, Cambridge university press, 2022.
2. Parkhurst, D.L., and Appelo, C.A.J., 2013, **Description of input and examples for PHREEQC version 3—A computer program for speciation, batch-reaction, one-dimensional transport, and inverse geochemical calculations**: U.S. Geological Survey Techniques and Methods, book 6, chap. A43, 497 p., <https://doi.org/10.3133/tm6A43>

References

1. Gustafsson, J.P., **Visual MINTEQ 3.0 user guide**, KTH, Department of Land and Water Resources, Stockholm, 2011.
2. Appelo and Postma, **Geochemistry, Groundwater, and Pollution**.
3. Merkel and Planer-Friedrich, **Groundwater Geochemistry**.
4. **Microsoft Excel**
5. USGS PHREEQC, open source (<https://www.usgs.gov/software/phreeqc-version-3>)
6. Visual MINTEQ, open source (<https://vminteq.lwr.kth.se/>)
7. Geochemist's Workbench, commercial (<https://www.gwb.com/>)
8. GWB Academy: <https://academy.gwb.com/academy.php>.